Name: SHEIKH WASIMUDDIN
Roll No: IOT-73
Branch: CSE(IOT)

LAB 07	
Aim:	Apply the concept of Files in java.
Theory:	In Java, with the help of File Class, we can work with files. This File Class is inside the java.io package. The File class can be used to create an object of the class and then specifying the name of the file. Why File Handling is Required? • File Handling is an integral part of any programming language as file handling enables us to store the output of any particular program in a file and allows us to perform certain operations on it. • In simple words, file handling means reading and writing data to a file. There are two standard methods to create a new file, either directly with the help of the File class or indirectly with the help of the FileOutputStream class by creating an object of the file in both approaches. Methods to Create Files in Java There are two methods mentioned below • Using the File Class • Using the FileOutputStream Class FileWriter class in Java is used to write character-oriented data to a file as this class is character-oriented because it is used in file handling in Java. There are many ways to write into a file in Java as there are many classes and methods which can fulfill the goal as follows: 1. Using writeString() method 1. Using FileWriter Class 1. Using BufferedWriter Class 1. Using FileOutputStream Class FileReader in Java is a class in the java.io package which can be used to read a stream of characters from the files. Java IO FileReader class uses either specified charset or the platform's default charset for decoding from bytes to characters. 1. Charset: The Charset class is used to define methods for producing encoders and decoders and for recovering several names combined with the charset.

Name: SHEIKH WASIMUDDIN
Roll No: IOT-73
Branch: CSE(IOT)

	2. Default Charset: The default charset is defined during implicit computer start-up and it depends on the locale and charset of the underlying operating system.

Program:	// sheikh wasimuddin 23071044 cseiot-73
	import java.io.File;
	import java.io.FileWriter;
	import java.io.IOException;
	
	public class Exp_7 {
	  public static void main(String[] args) {
	  try {
	  // Create a new file
	  File myFile = new File("example.txt");
	  if (myFile.createNewFile()) {
	  System.out.println("File created: " + myFile.getName());
	  } else {
	  System.out.println("File already exists.");
	  }
	
	  // Write to the file
	  FileWriter writer = new FileWriter("example.txt");
	  writer.write("Hello, my name is sheikh wasimuddin.");
	  writer.close();
	  System.out.println("Successfully wrote to the file.");
	
	  } catch (IOException e) {
	  System.out.println("An error occurred.");
	  e.printStackTrace();
	  }
	  }
	}

Output:	

Name: SHEIKH WASIMUDDIN
Roll No: IOT-73
Branch: CSE(IOT)

Exp_7.java 2 ✕ example.txt

Exp_7.java > ...

```java
// sheikh wasimuddin 23071044 cseiot-73
import java.io.File;
import java.io.FileWriter;
import java.io.IOException;

public class Exp_7 {
    Run main | Debug main | Run | Debug
    public static void main(String[] args) {
        try {
            // Create a new file
            File myFile = new File(pathname:"example.txt");
            if (myFile.createNewFile()) {
                System.out.println("File created: " + myFile.getName());
            } else {
                System.out.println(x:"File already exists.");
            }

            // Write to the file
            FileWriter writer = new FileWriter(fileName:"example.txt");
            writer.write(str:"Hello, my name is sheikh wasimuddin.");
            writer.close();
            System.out.println(x:"Successfully wrote to the file.");

        } catch (IOException e) {
            System.out.println(x:"An error occurred.");
            e.printStackTrace();
        }
    }
}
```

PROBLEMS 4 OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\student\Desktop\Iot-73> & 'C:\Program Files\Java\jdk-22\bin\java.exe' '-XX:+ShowCodeDetailsI
 'Exp_7'
File created: example.txt
Successfully wrote to the file.
PS C:\Users\student\Desktop\Iot-73>
```

Exp_7.java 2 example.txt ✕

example.txt

```
1   Hello, my name is sheikh wasimuddin.
```

Conclusion	Implementing the concept of Files in java executed successfully.
Name Of Faculty:	Prof. Akhil Jajulwar